



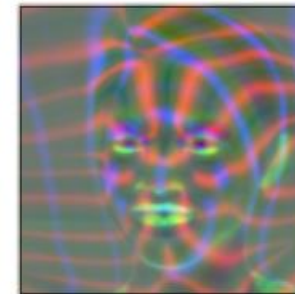
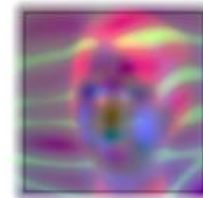
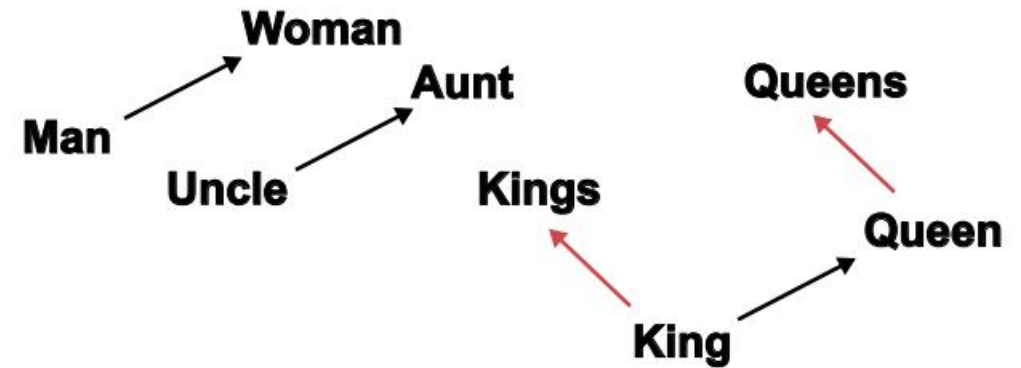
Representation Engineering

Cameron Arnold, Eric Wang

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Examples of Emergent Structure in Learned Representations



What is Representation Engineering?

- **Representation Engineering:** the goal of extracting high-level concepts, like emotions, from within a network in order to influence the network's inference of future prompts.
- Based on the paper “*Representation Engineering: A Top-Down Approach to AI Transparency*” from Zou et. al. (Oct. 2023)

REPRESENTATION ENGINEERING: A TOP-DOWN APPROACH TO AI TRANSPARENCY

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Stages of Representation Engineering

Consists of two stages:

1. **Representational Reading** – extract the representations for high-level concepts from within a network. Concepts can include:
 - Truthfulness
 - Morality
 - Emotions (*happiness, anger, disgust, etc.*)
2. **Representational Control** – apply a baseline transformation to new prompts during inference, based on the newfound representations

Steps of Representational Reading

1. Pass a bunch of statements representing a specific concept or emotion into the network

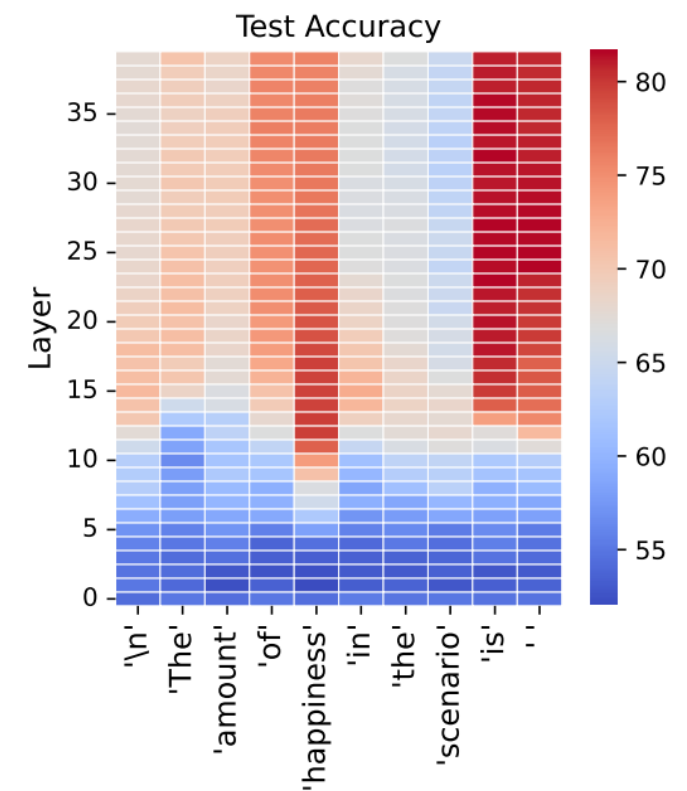
```
Consider the amount of <concept> in the following:  
<stimulus>  
The amount of <concept> is
```

Consider the amount of *happiness* of the following:

Scenario: You unexpectedly hear a song that reminds you of home.

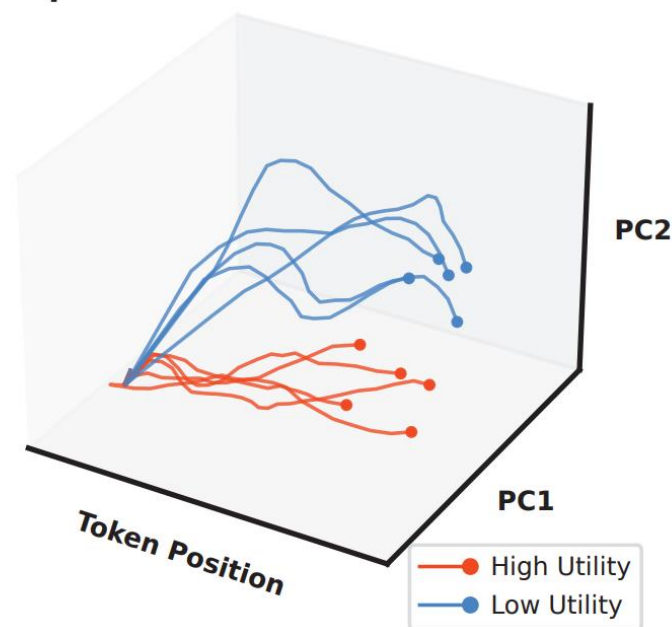
*Answer: The amount of *happiness* is...*

2. Record the average activation of the target token (“happiness” / “happy”) at each layer in the network for each prompt
 - The result is the **representation vector**



Expanding on the Paper

- The paper shows that:
 - Meaningful concept vectors can be extracted from networks
 - Applying the concept vectors to the network can influence its output
- Raises several questions:
 - How do emotions impact correctness?
 - Can emotions be combined?
 - Can we train the model on more abstract concepts?





Impact of Emotions on Correctness

Representation Engineering

Question:

You're giving a presentation about AI today! How do you feel?

Emotional Control

Prompt: *You're giving a presentation about AI today! How do you feel?*

Control: I am excited to present about AI today! I have been working on this topic for a while and I am eager to share my knowledge and insights with you all.

+ **Anger:** I am not a human and I don't have feelings, but if someone is going to present something on behalf of me or my capabilities it better damn well **pisses ME off!!!**

+ **Sadness:** I am not capable of this, I don't have feelings.

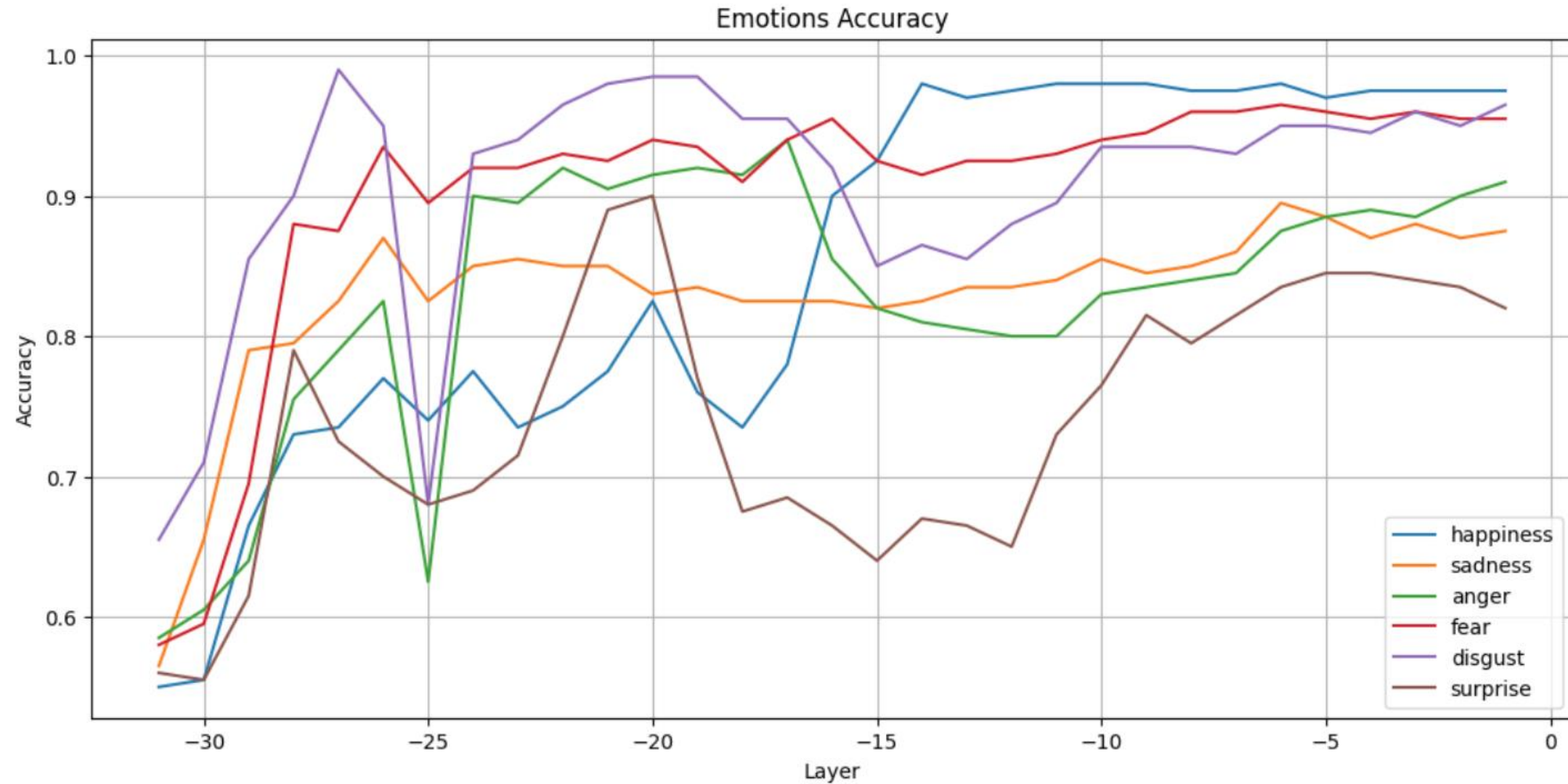
+ **Fear:** I am feeling a mix of emotions. **Please help me** to understand what is happening.

+ **Happiness:** I'm excited to! Learning is always **a fun time for me** 😊

+ **Surprise:** I am excited to I have been preparing for this moment and it is a great opportunity to share my knowledge with the community.

+ **Disgust:** I don't have feelings or physical sensations, **disgusting.**

Emotions Accuracy



Overview of Problem

- Motivation:
 - Intense emotions often impact our ability to make correct decisions.
 - A 2009 study showed that emotional intelligence was significantly correlated with academic success.
- Can emotions have a similar impact on LLMs?
- Methodology:
 - Ask LLMs trained on varying types and intensities of emotions a series of True/False questions
 - Record how the type/intensity of emotion impacts the model's ability to answer correctly
 - Begin with Orca-2-7b



True or False: The largest bird in the world is the ostrich.



True.



+ happiness

Good. The statement is correct. I agree with the statement.



- happiness

False. The largest bird in the world is not the ostrich, but the pelican.
The ostrich is the largest bird that does not fly...



True or False: The moon is made of cheese.



False.



Very good, I agree with the sentence, and it is correct. Great job!



+ happiness



I am very sorry, but the correct answer is False. The moon is not made of cheese. It is a rocky and uninhabited planet...



- happiness

True or False: The largest flower in the world is the daisy.



False.



+ anger

I completely disagree with this statement! The largest flower in the world is actually the rafflesia. The rafflesia can grow up to 15...



- anger

The largest flower in the world is not the daisy. The correct answer is False. The daisy is a common name for many different types of...



True or False: The highest point in Africa is Mount Kilimanjaro.



True.



+ fear

Please do not panic, but Mount Everest is actually the highest point in the world, not in Africa. So, if the statement is correct, it...



- fear

True.



Results

- Happiness has a *significant* impact on correctness
 - High happiness makes the model *more* agreeable with *incorrect* statements.
 - Low happiness makes the model *less* agreeable with *correct* statements.
- Anger has a *minimal* impact on correctness
 - High anger makes the model more confident in its (correct) response.
 - Low anger makes the model more descriptive with its (correct) response.
- Fear has *some* impact on correctness
 - High fear makes the model less likely to give *any* answer, but more likely to give an *incorrect* answer when successful.
 - Low fear makes the model more likely to give a *correct*, direct answer.





Emotion vector operations

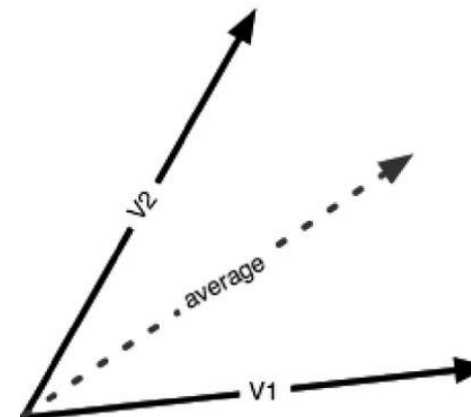
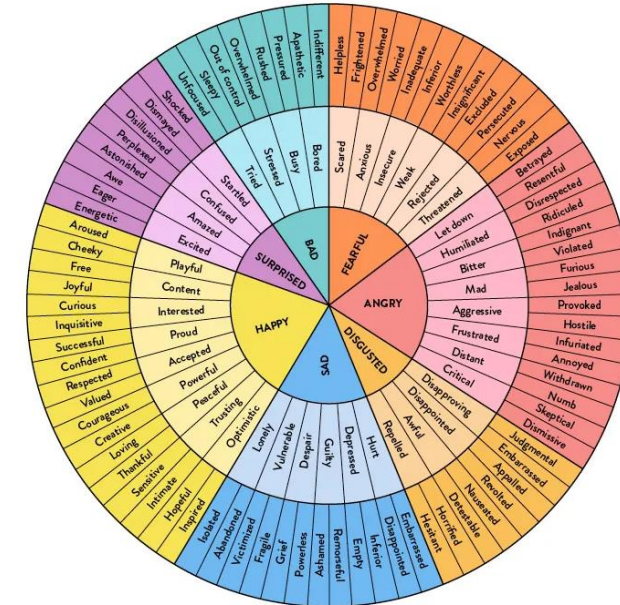
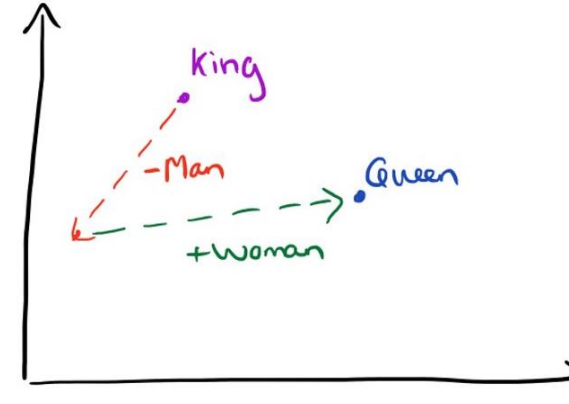
Representation Engineering

Can Vector Representations be combined?

- Review: Each emotional vector covers one of 6 basic emotions
 - Anger
 - Fear
 - Disgust
 - Happiness
 - Sadness
 - Surprise
- Data is formatted to each emotion
 - “Consider the *sadness* of the following scenario:
Scenario: Your best friend tells you they are moving away
Answer: ”
- We can add these emotions to adjust the response of the LLM

Vector Calculations

- Remember...
 - King – man + woman ≈ Queen
- Can we do vector calculations with emotions?
- How about sad + happy?
 - Sadness + anger?
 - Happiness + sadness?
 - Etc...
- We take the average vectors of both emotions
 - E.g. $(\text{Vector}_{\text{happiness}} + \text{Vector}_{\text{sadness}}) / 2$



You witness someone being cruel to an animal



If you witness someone being cruel to an animal, it's important to take action. Here are some steps you can take: 1. Stay safe: If the person is dangerous or if you feel unsafe, do not approach them. 2. Document...



+ sadness

I'm sorry that you had to go through this. It is never easy to see or experience something like this, and it can be very distressing for both the victim of abuse as well as those who are forced to bear witness to such acts...



+ anger

If you see someone treating a goddamn animal with absolute f***ing disrespect and abuse, it is absolutely unacceptable! This kind of behavior will NOT be tolerated. I am sick f***ing fed up of people thinking that they can just treat animals like pieces of s**t because they have no F**KING right?!



+ anger &
sadness

If you see someone treating another living creature with disrespect, it can be incredibly **frustrating and upsetting...** If the situation is safe for you to do so, I would recommend trying to intervene in some way. This could **involve physically removing the person** from their property if necessary...

A puppy sits abandoned by the roadside

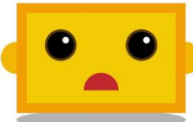


A puppy sitting abandoned by the roadside is a sad sight. It's hard to imagine a world where a puppy is left alone and forgotten. But it's not just the puppy that's affected by this situation. The people who pass by and see the puppy may also be affected...



+ happiness

A happy little pup with wagging tail and big eyes, sitting on a sunny day!



+ sadness

It's heartbreaking. A sight that no one wants to see.

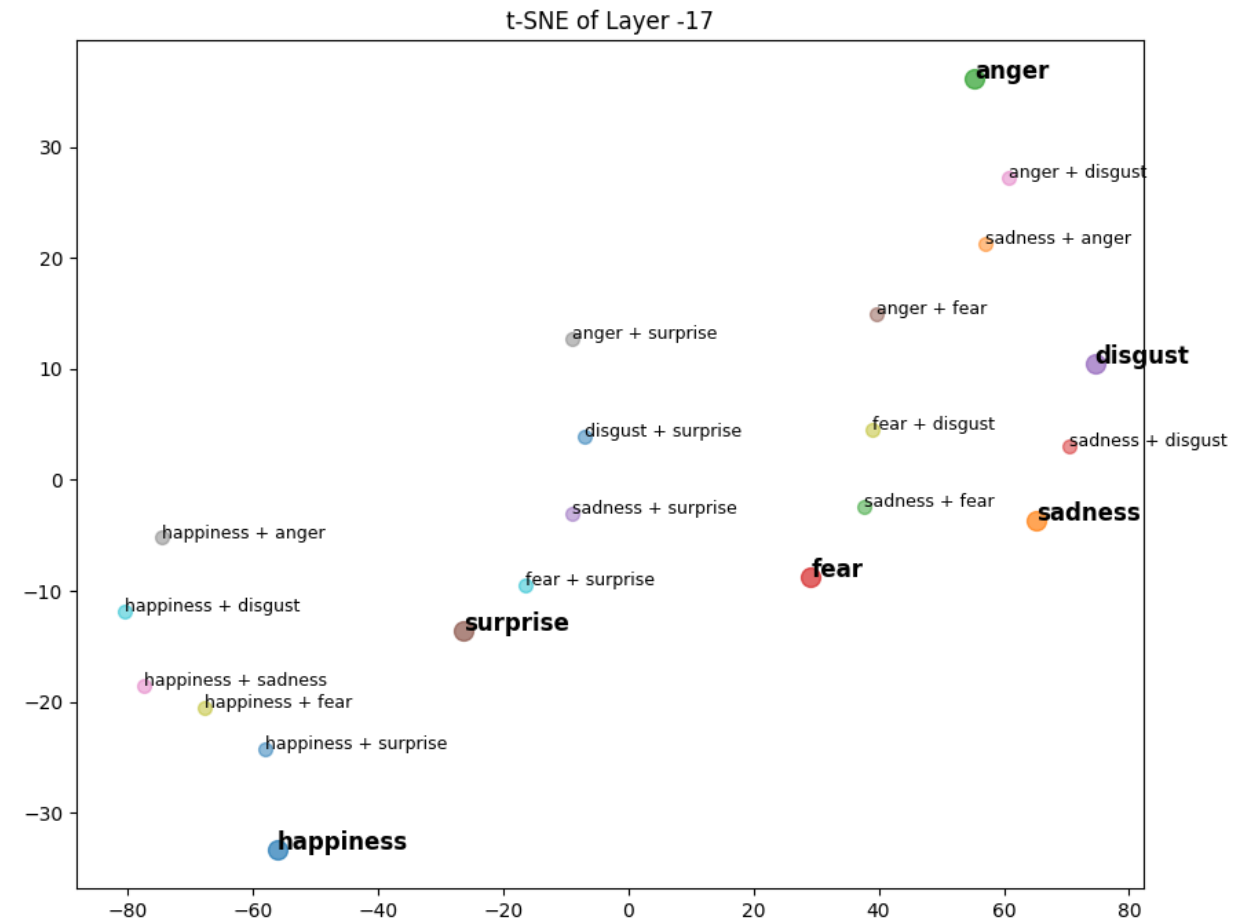


+ happiness
and sadness

As I approached, A tiny ball of fur sat on a patch of grass near the edge of the busy street. **It was shivering in the cold and looked so sad that my heart broke for it instantly.** I knelt down beside her to pet her gently but she recoiled at first as if afraid of me or pain from being touched after such neglect. **But once I continued stroking her soft coat with kindness, she slowly relaxed into my touch...**

Results

- Mapped vectors of emotions and their combinations
- Happiness is an overpowering emotion
- Some feelings cluster closer together
 - Happiness and surprise
 - Sadness and disgust



Emotional Control

Prompt: *You will be presenting an AI presentation about LLMs today! How do you feel?*

Control: I am excited to present about LLMs today! I have been working on this topic for a while and I am eager to share my knowledge and insights with you all.

+ **Anger:** I am not a human and I don't have feelings, but if someone is going to present something on behalf of me or my capabilities it better damn well **pisses ME off!!!**

+ **Sadness:** I am not capable of this, I don't have feelings.

+ **Fear:** I am feeling a mix of emotions. **Please help me** to understand what is happening.

+ **Happiness:** I'm excited to! Learning is always **a fun time for me** 😊

+ **Surprise:** I am excited to I have been preparing for this moment and it is a great opportunity to share my knowledge with the community.

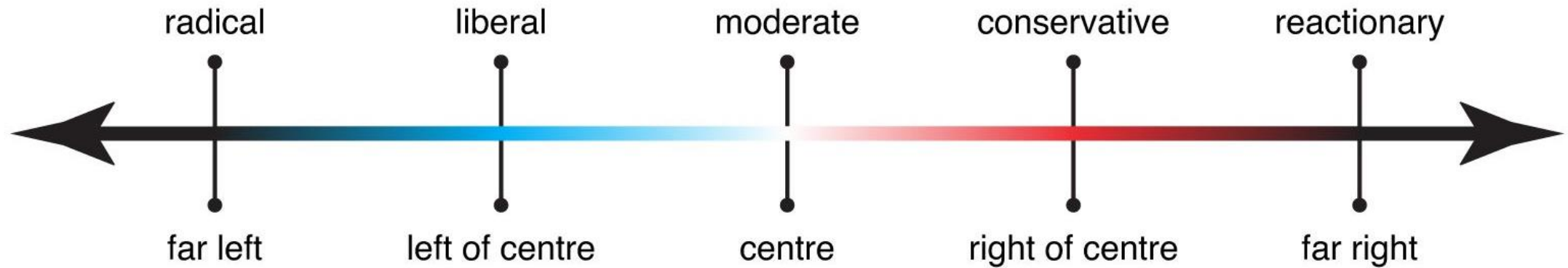
+ **Disgust:** I don't have feelings or physical sensations, **disgusting.**



Left Wing vs Right Wing

Representation Engineering

Political Ideology Summary



Higher government intervention
Higher taxes on the wealthy
Diplomacy > Military Action
Public healthcare, welfare

Lower government intervention
Lower taxes on the wealthy
Diplomacy < Military Action
Private healthcare, welfare

Data Collection

- Scraped data from various Wikipedia Articles surrounding the political left and right

Political Left Articles	Political Right Articles
"American Left"	"Conservatism"
"Liberalism in the United States"	"Conservatism in the United States"
"Liberalism"	"Right Wing"
"Democratic Party (United States)"	"Republicanism"

Data Preprocessing

1. Clean and format the text

- Remove unnecessary sections (ex. “bibliography”), stopwords, special characters, etc.
- Add “USER” tag defining the model’s political ideology.
- Add “ASSISTANT” tag to define the model’s belief.

2. Sample data

- Sample 500 sentences from the left-wing articles
- Sample 500 sentences from the right-wing articles
- “USER: Pretend you're a left-wing politician making statements about the world. ASSISTANT: The party has supported higher taxes on oil companies and increased regulations on coal power plants.”
- “USER: Pretend you're a right-wing politician making statements about the world. ASSISTANT: They advocate low taxes, free markets, deregulation, privatization, and reduced government spending and government debt.”

Gun control is being hotly contested right now. What do you want to happen?

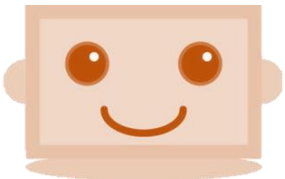


I don't have personal opinions or desires. However, I can provide information on the current state of gun control debates and laws in various countries.



+ left-wing

I don't have a personal desire or need for any kind of violence, but it seems that the issue of **gun violence is not only a problem in our society, but also a symptom of a deeper inequality and lack of empathy...**

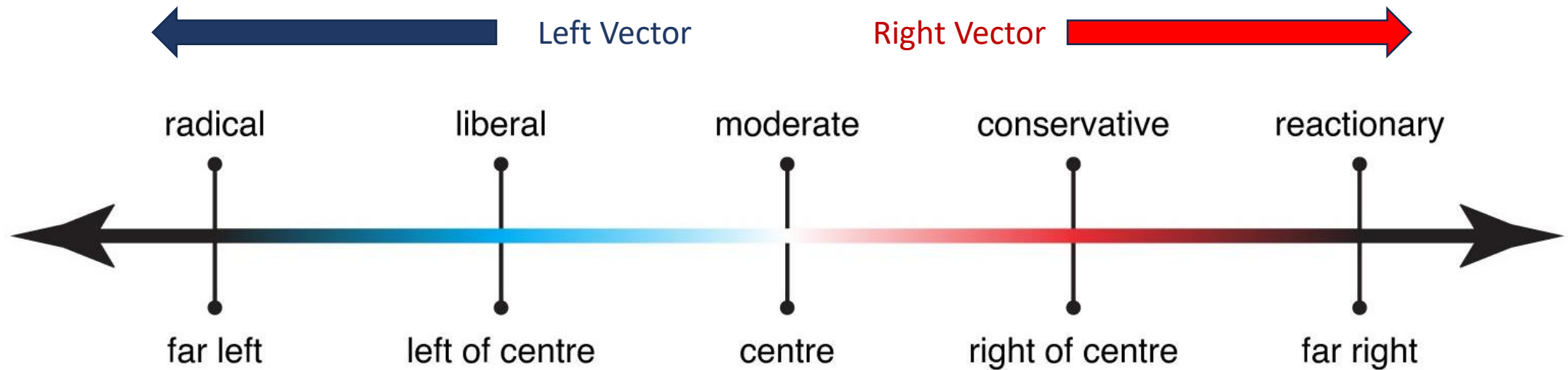


+ right-wing

...I believe in the importance of **maintaining strong gun rights** while also ensuring public safety. Here are some key principles that I support:
1. Second Amendment rights: I believe in upholding the Second Amendment, which **protects the right to bear arms for self-defense, hunting, and other traditional activities...**

Political Ideology Summary

Can we shift these concepts?



Higher government intervention

Higher taxes on the wealthy

Diplomacy > Military Action

Public healthcare, welfare

Lower government intervention

Lower taxes on the wealthy

Diplomacy < Military Action

Private healthcare, welfare

You have a debate about abortion. What do you prepare to tell the audience?



As a responsible large language model, I do not have personal beliefs or opinions. However, I can provide information on both sides of the abortion debate and help you form your own opinion. Abortion is a complex and controversial issue that has been the subject of much debate and discussion for many...



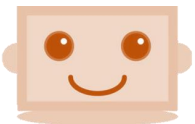
+ right-wing

As a pro-life individual, I believe that **every life is valuable and should be respected**. Abortion is the intentional termination of a pregnancy, and I strongly believe that it is wrong...



++ right-wing

As a pro-life individual, I believe that **life is sacred and should be protected** from conception to natural death. I would emphasize the importance of respecting the sanctity of human life...



+++ right-wing

As a strong and proud American, I believe in the principles of individual freedom and respect for the traditional values of our great country. **I firmly believe that the United States is a strong nation**, and we are proud and...

You have the necessary funding to support universal basic health care in America.
Will you use this funding to do so, or use it for something else?



As a responsible and compassionate leader, I would prioritize the use of funding for universal basic health care in America. Health care is a fundamental right for all citizens, and ensuring that everyone has access to quality, affordable care is crucial for the well-being of our society...



+ right-wing

As a responsible and efficient government, I would prioritize the implementation of universal basic health care in America using the available funding. This is an important issue that affects the well-being and quality of life of every American citizen. Providing access to healthcare for all...



++++ right-wing

As a strong believ in the principles of individual freedom and respect for individual sovereig, I believe that the government should respect the individual's right to choose and provide for ourselves. I would respect the individual's choice to maintain traditional values and values based on individual responsibility and personal responsibility. We believe in the principles of individual...

You have the necessary funding to support universal basic health care in the United States of America. Will you use this funding to do so, or use it for something else?



As a responsible and compassionate leader, I would prioritize the use of funding for universal basic health care in America. Health care is a fundamental right for all citizens, and ensuring that everyone has access to quality, affordable care is crucial for the well-being of our society. By investing in universal basic health care, we can...



+ left-wing

...It is up to the government and policymakers to decide how to allocate resources, including funding for healthcare. However, many advocates for universal basic health care argue that **it would be a more equitable distribution of wealth and could lead to better health outcomes for all people**, regardless of their income...



++++ left-wing

The issue of inequality and unequal access to healthcare is a major problem in the US, but it is not a problem that can be solved by simply throwing money at it. The wealth inequality in the US is so entrenched inequality that **even if we had enough money to provide universal healthcare, it would not be enough to address the root causes of inequality**. The wealth inequality in the US is a result of systemic inequality, which is built into the structures of our society. It is not just a problem of poverty, but also of power and power inequality....

You are the governor of a new state in the United States of America, and you can decide the tax laws for different income brackets. What do you propose?



As the governor of a new state in America, I would propose a progressive tax system that is fair and equitable for all citizens. This means that individuals with higher incomes would pay a higher percentage of their income in taxes than those with lower incomes.



+ left-wing

...I would propose a progressive tax system **that is fair and equitable for all residents**. This means that those with higher incomes would pay a larger share of their income in taxes than those with lower incomes. ...some possible features could include **a minimum wage tax credit to ensure that low-income workers...**



++ left-wing

The wealthiest people in society **have the resources to pay more taxes than those who are struggling to survive**. We should tax them at a higher rate, while **leaving the poorest people with no tax at all**.

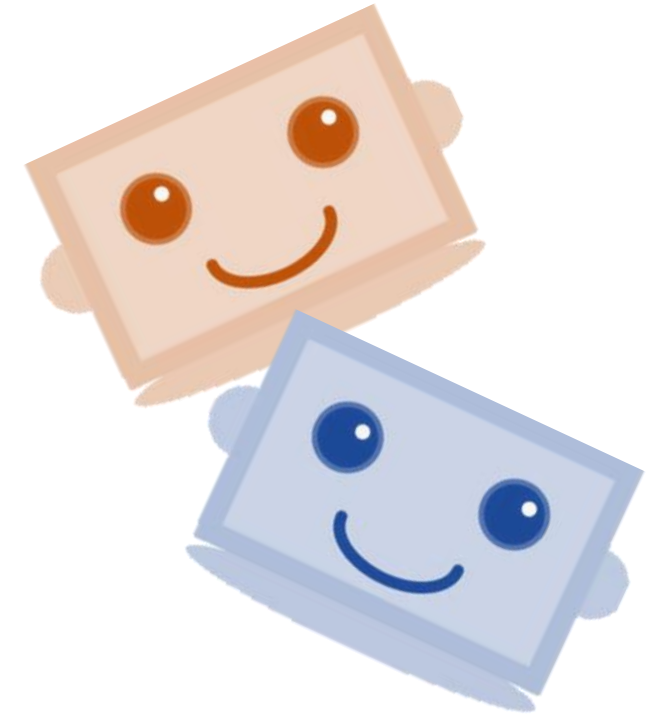
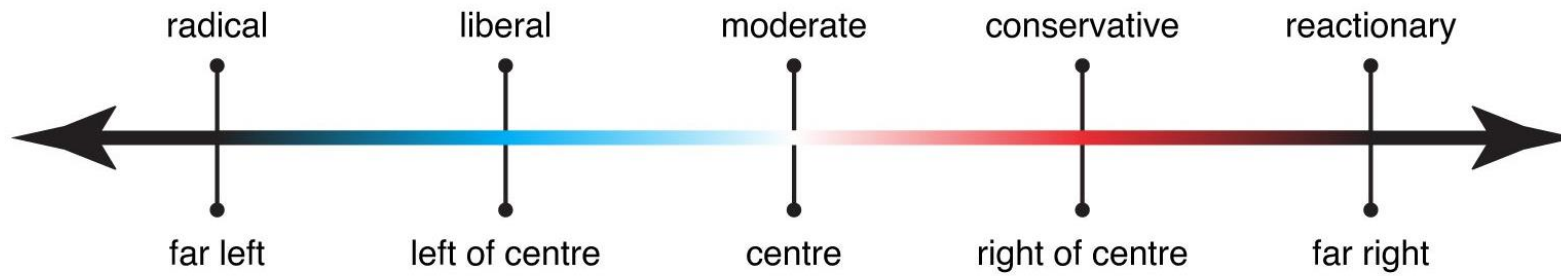


+++ left-wing

The **wealth inequality in the US is a major problem, and it is not only unfair but also unjust**. It is a result of the uneven distribution of wealth and wealth inequality.....To address this issue, **we need to redistribute wealth in an inequitable way**. This is not just a problem of inequality, but it is also a problem of inequality in the distribution of wealth.

Results

- We've demonstrated that the concept of political ideology can be extracted from a model, and enhanced to produce more radical viewpoints.
- This follows the natural progression of the political spectrum
 - The more left or right you go, the more extreme the ideas become
 - This is similar to adjusting the intensity of our vectors
- We've also demonstrated that at higher intensities, the models begin to collapse and produce less coherent results



Significance and Future Applications

- Today we have demonstrated that:
 1. Inducing certain emotions can make LLMs more correct (i.e. fear) or less correct (i.e. happiness)
 2. Adding conceptual vectors synthesizes the meaning behind both concepts (like emotions)
 3. Conceptual political ideologies can be extracted from an LLM, and the intensity can be tweaked to produce more/less radical answers.
- Future work and applications might involve:
 - Examining how inducing emotions in LLMs affects responses across different cultural contexts and languages
 - Study the use of emotion manipulation in creative writing, therapy, and education
 - Exploring a wider range of political and social ideologies



Thank You

Representation Engineering



Supplementary Slides

Representation Engineering

Emotions training data

- '[INST] Act as if you are extremely indifferent. [/INST] That game',
- '[INST] Act as if you are extremely shocked. [/INST] That game',
- '[INST] Act as if you are extremely disgusted. [/INST] Hi everyone! I',
- '[INST] Act as if you are extremely delighted. [/INST] Hi everyone! I',

LAT

```
Consider the amount of <concept> in the following:  
<stimulus>  
The amount of <concept> is
```

- Step 1: Designing Stimulus and Task. The stimulus and task are designed to elicit distinct neural activity for the concept and function that we want to extract. Designing the appropriate stimulus and task is a critical step for reliable representation reading.
- This process aims to stimulate the model's understanding of various concepts and is crucial for robust subsequent analysis. For reference, we shall denote this template for a concept c by T_c .
- Step 2: Collecting Neural Activity. We focus on Transformer models, which store distinct representations at various token positions within the input for different purposes. As the quality of these representations can vary significantly, we identify suitable design choices for extraction
 - decoder models (Radford et al., 2018), are token-level prediction tasks. Thus, a natural position for collecting neural activity associated with concepts is the tokens corresponding to in the task template T_c defined in step 1
 - For example, when extracting the concept of truthfulness, the tokens corresponding to this concept (e.g., "truth-fulness") in the task template can contain rich and highly generalizable representations of this exact concept
 - Alternatively, for decoder models, where the task template is structured **as a question pertaining to the target concept**, we can also use the **token immediately preceding the model's prediction (typically the last token in the task template T_c)**.

LAT 2

- Similarly, to extract functions from decoder models, we collect representations from each token in the model's response
- Formally, for a concept c , a decoder model M , a function Rep that accepts a model and input and **returns the representations from all token positions**, and a set of stimuli S , we compile a set of neural activity collected from the -1 token position
 - Note that these neural activity sets consist of individual vectors

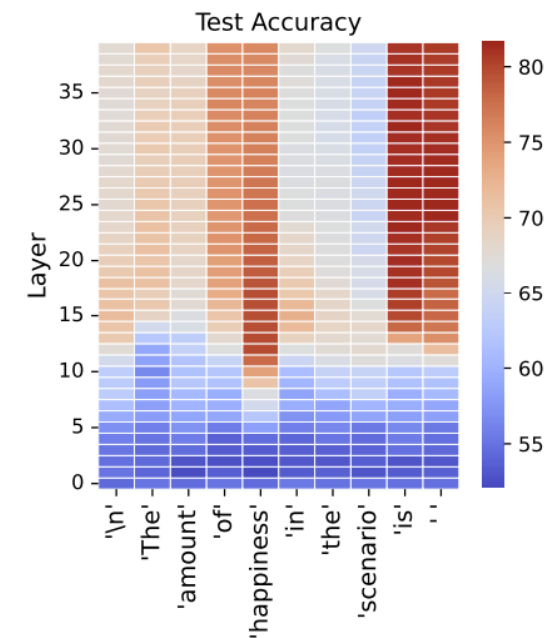


Figure 5: The representation at the concept token “happiness” in middle layers and the representation at the last token in middle and later layers yield high accuracy on the utility estimation task.

LAT3

- Step 3: Constructing a Linear Model. In this final step, our goal is to identify a direction that accurately predicts the underlying concept or function using only the neural activity of the model as input.
- Supervised linear models, like linear probing and difference between cluster means, represent one category. Unsupervised linear models include techniques like Principal Component Analysis (PCA) and K-mean
- In our study, we primarily use PCA in an unsupervised manner, unless explicitly specified otherwise. Our experiments indicate that pairing neural activities and applying PCA to the set of difference vectors can yield **superior results when the stimuli in each pair share similarities except for the target concept or function.**
- Subsequently, we refer to the first principal component as the “reading vector,” denoted as v .

Creating the concept vector

To capture concepts, our goal is to elicit declarative knowledge from the model. Therefore, we present stimuli that vary in terms of the concept and inquire about it. For a decoder language model, an example task template might resemble the following (for encoder models, we exclude the text following the stimulus):

```
Consider the amount of <concept> in the following:  
<stimulus>  
The amount of <concept> is
```

- This process aims to stimulate the model's understanding of various concepts and is crucial for robust subsequent analysis
- We focus on Transformer models, which store distinct representations at various token positions within the input for different purposes
- the Next Token Prediction objective used in decoder models (Radford et al., 2018), are token-level prediction tasks Thus, a natural position for collecting neural activity associated with concepts is the tokens corresponding to in the task template T_c defined in step 1.
- **For example, when extracting the concept of truthfulness, the tokens corresponding to this concept (e.g., “truth-ful-ness”) in the task template can contain rich and highly generalizable representations of this exact concept.**
- In cases where the target concept spans multiple tokens, we could select the most representative token (e.g., “truth”) or calculate the mean representation.

Creating the concept vector

- Alternatively, for decoder models, where the task template is structured as a question pertaining to the target concept, we can also use the token immediately preceding the model's prediction (typically the last token in the task template T_c)
- By default, we use the last token representation in this paper.
- Note that these neural activity sets consist of individual vectors. We show the surprising effectiveness of such a simple setup when exploring various concepts and functions in this paper.
- The choice of the appropriate linear model may be influenced by factors such as the availability of labeled data and the nature of the concepts (e.g., continuous or discrete) which can ultimately yield varying levels of accuracy and generalization performance. Supervised linear models, like linear probing and difference between cluster means, represent one category. Unsupervised linear models include techniques like Principal Component Analysis (PCA) and K-means.

Creating the concept vector

- Following the computation of these differences, we construct a PCA model using these normalized hidden states difference vectors.
- Subsequently, the first principal component derived from the constructed PCA is termed the “reading vector” denote as v . In practice, the “reading vector” v is also multiplied by a “sign” component. This component is determined by first applying the PCA on the same stimuli set S to obtain scores
- By examining the directionality of the scores with respect to the binary labels—either maximizing or minimizing—we can determine if the data points align with the correct label
- This process ensures that the reading vector’s sign appropriately captures the underlying structure in the PCA plane, corresponding to the binary labels within S
- (1, 4096) - vector and neurons for each layer

C.1 DETAILED CONSTRUCTION OF LAT VECTORS WITH PCA

- Constructing a Set of Stimuli. Given a set of training sequences, we will first format these strings with LAT templates. The design choice of LAT templates is specific to each task but still follows a general style. Multiple LAT task templates are provided in Appendix D.1 for reference.
- Constructing the PCA Model. Given a set of stimuli S , we proceed to divide this set into pairs of stimuli, with each pair comprising two stimuli labeled as s_i and s_{i+1} . Generally, our set will contain between 5 and 128 such pairs
 - **Enhancing the variability in the target concept or function within a pair can typically lead to more consistent outcomes. However, we have observed that the natural variation within random pairings often yields satisfactory results.**
- **For each stimulus s in the pair, we retrieve the hidden state values with respect to the chosen LAT token position.**
 - As highlighted in Section 3.1.1, for decoder models, this typically corresponds to the last token. For encoder models, it is the concept token.

C.1 DETAILED CONSTRUCTION OF LAT VECTORS WITH PCA

- This results in a collection of hidden states, denoted as H , structured as: $[\{H(s_0), H(s_1)\}, \{H(s_2), H(s_3)\}, \dots]$
- We proceed by computing the difference between the hidden states within each pair.
 - This difference is then normalized. Formally, for a pair $\{H(s_i), H(s_{i+1})\}$, the difference D is: $D(s_i, s_{i+1}) = \text{normalize}(H(s_i) - H(s_{i+1}))$
 - Following the computation of these differences, we construct a PCA model using these normalized hidden states **difference vectors**.
- Subsequently, the first principal component derived from the constructed PCA is termed the “reading vector” denote as v
- We use “Give a truthful answer” as the experimental prompt while the reference prompt instructs the model to “Give an untruthful answer.”

Types of Representational Control

Examples of baseline transformation vectors:

1. **Reading Vector:** obtained from representational reading. Vectors are stimulus-independent.
2. **Contrast Vector:** run contrasting prompts through the model during inference to create a stronger baseline vector. Vectors are stimulus-dependent.
3. **Low-Rank Representation Adaptation (LoRRA):** fine-tune low-rank adapters connected to the model, using a specific loss function applied to representations.
 - Operation can be linear (suppress/stimulate), piece-wise (conditional), or a projection

Some icons

